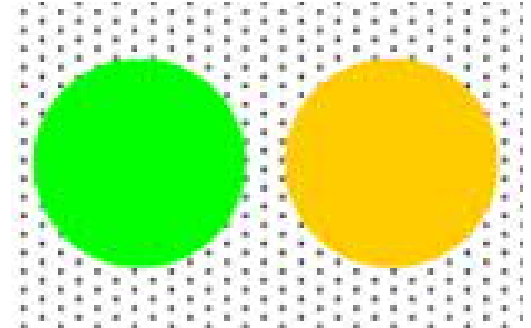
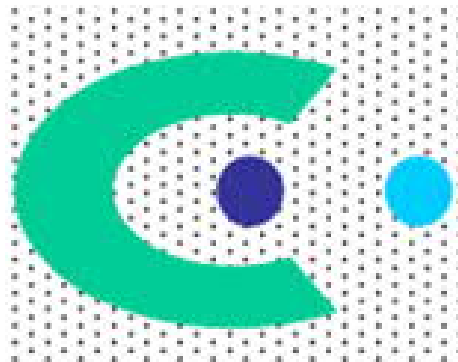


Outline

- ▶ What is Clustering?
- ▶ Types of Data in Cluster Analysis and Similarity Measures
- ▶ Some clustering Methods
 - ▶ K-means and K-means++
 - ▶ K-medoids
 - ▶ Hierarchical clustering method
 - ▶ DBSCAN: a Density-based Algorithm
- ▶ Cluster Validity

Density-based Clustering

- ▶ A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
- ▶ Used when the clusters are **irregular** or intertwined, and when noise and outliers are present.



Density-based Clustering

- ▶ Clustering based on density (local cluster criterion), such as density-connected points
- ▶ Major features:
 - ▶ Discover clusters of arbitrary shape
 - ▶ Handle noise
 - ▶ Do not need to specify k , but need density parameters
- ▶ Several interesting studies:
 - ▶ DBSCAN: Ester, et al. (KDD'96)
 - ▶ OPTICS: Ankerst, et al. (SIGMOD'99)
 - ▶ DENCLUE: Hinneburg & D. Kein (KDD'98)
 - ▶ CLIQUE: Agrawal, et al. (SIGMOD'98)

DBSCAN: Basic Concepts

- ▶ ***Eps***-neighborhood of point p in data set D :

$$N_{Eps}(p) = \{q \in D \mid \text{dist}(p, q) \leq Eps\}$$

where ***Eps*** is called the radius of the neighborhood

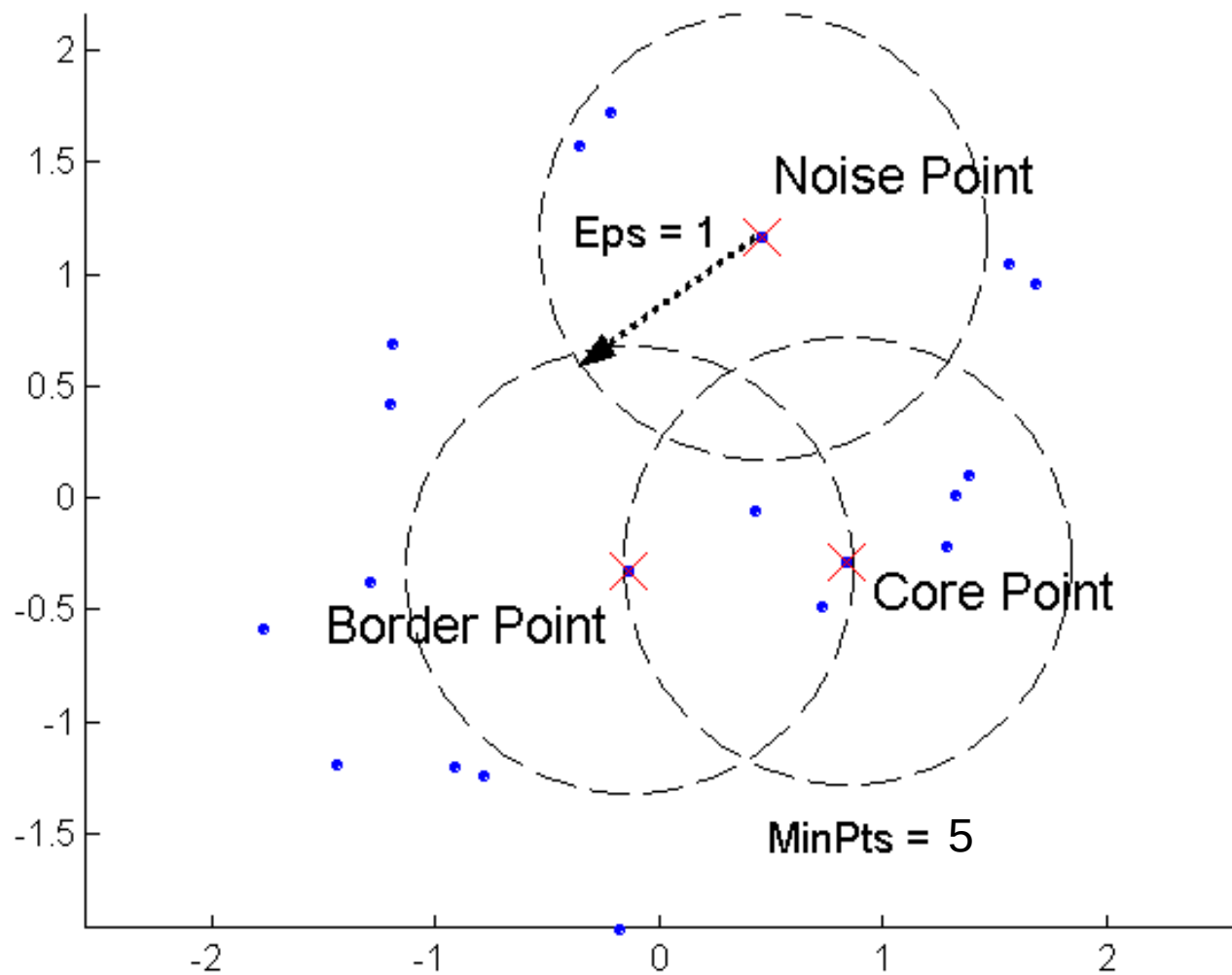
- ▶ Density of ***Eps***-neighborhood of p :

the number of points in $N_{Eps}(p)$

DBSCAN: Basic Concepts

- ▶ A point is a **core point** if there are at least *MinPts* number of points in its *Eps*-neighborhood.
 - ▶ These are points that are at the interior of a cluster
- ▶ A **border point** has fewer than *MinPts* points in its *Eps*-neighborhood, but is in the *Eps*-neighborhood of a core point.
 - ▶ These are points that are on or close to the border of a cluster
- ▶ A **noise point** is any point that is not a core point or a border point.
 - ▶ These are points that are outside any cluster

DBSCAN: Core, Border and Noise Points



DBSCAN: Input Parameters

- ▶ ***Eps:***

Maximum radius of the neighbourhood

- ▶ ***MinPts:***

Minimum number of points in an *Eps*-neighbourhood of a core point

DBSCAN: Basic Concepts

- ▶ Directly density-reachable:

- ▶ A point p is **directly density-reachable** from a point q wrt. Eps and $MinPts$ if

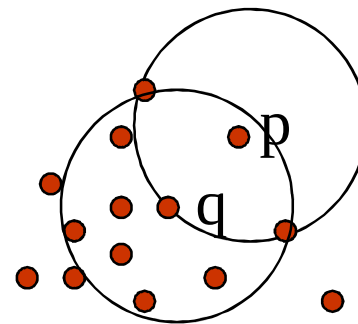
- 1) p belongs to $N_{Eps}(q)$

- 2) q is a core point, that is:

$$|N_{Eps}(q)| \geq MinPts$$

- ▶ Asymmetric in general

- ▶ Symmetric only when both p and q are core points.



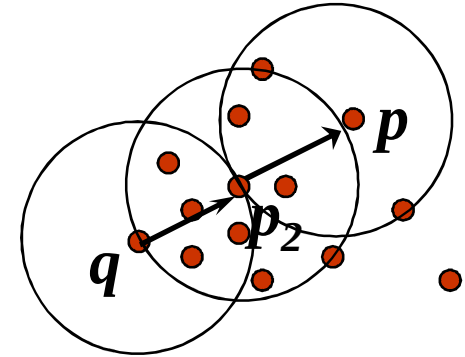
MinPts = 5

Eps = 1 cm

DBSCAN: Basic Concepts

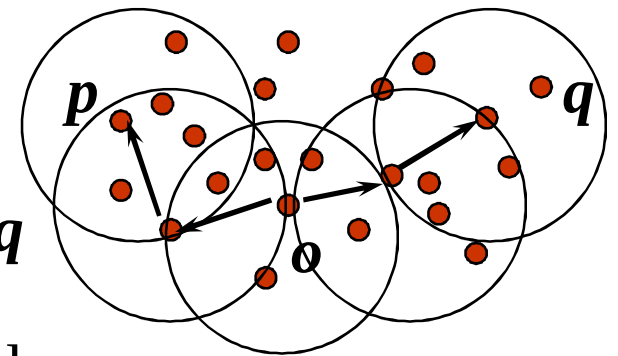
► Density-reachable:

- A point p is **density-reachable** from a point q wrt. Eps and $MinPts$ if there is a chain of points $p_1, p_2, \dots, p_n, p_1 = q, p_n = p$ such that p_{i+1} is directly density-reachable from p_i
- Symmetric only when p and q are both core points



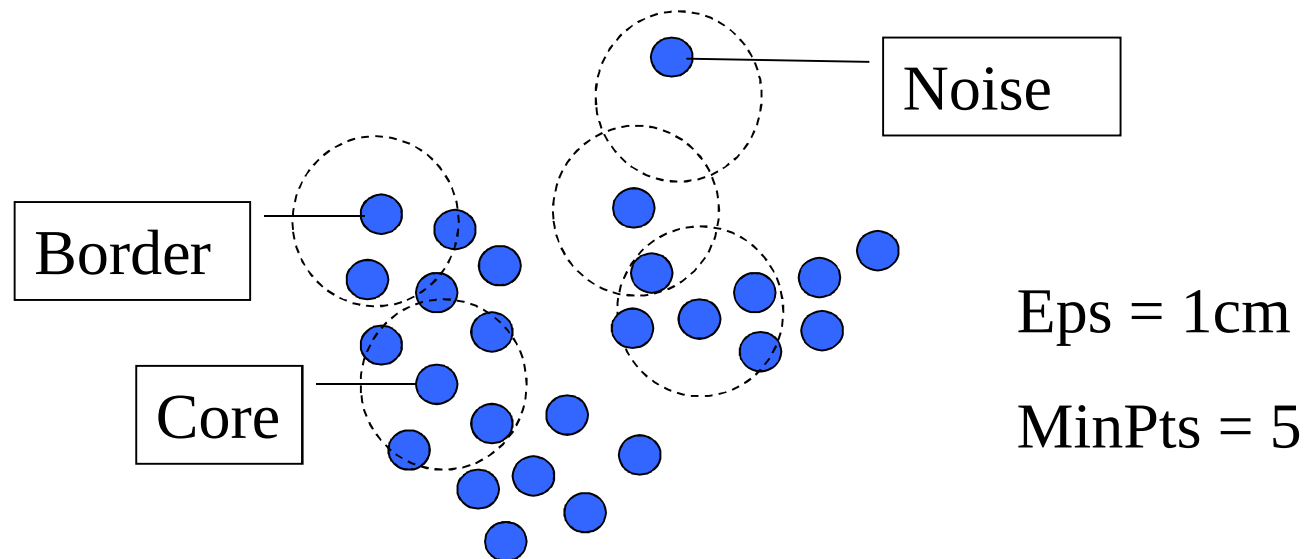
► Density-connected:

- A point p is **density-connected** to a point q wrt. Eps and $MinPts$ if there is a point o such that both p and q are density-reachable from o wrt. Eps and $MinPts$.
- Symmetric



DBSCAN: Cluster

- ▶ *Density-based notion of cluster*: A **cluster** is defined as a **maximal set of density-connected points**
 - ▶ Each pair of points in a cluster are density-connected to each other (Connectivity)
 - ▶ Core points in other clusters are not density-connected to any core points in this cluster (Maximality)
- ▶ Noise points are not in any cluster



DBSCAN: The Algorithm

- ▶ Arbitrarily select an unprocessed point p
- ▶ If p is a core point,
 - ▶ Retrieve all points density-reachable from p wrt Eps and $MinPts$.
 - ▶ A cluster is formed which includes all the points density-reachable from p
 - ▶ Mark all the points in the cluster as “processed”
- ▶ If p is not a core point, no points are density-reachable from p and DBSCAN visits the next unprocessed point of the database.
- ▶ Continue the process until all of the core points have been processed.

These points and p are in a cluster because they are density-connected through p .

Note: Membership of a border point depends on the order of the points being processed if it is density-connected to core points in two or more clusters.

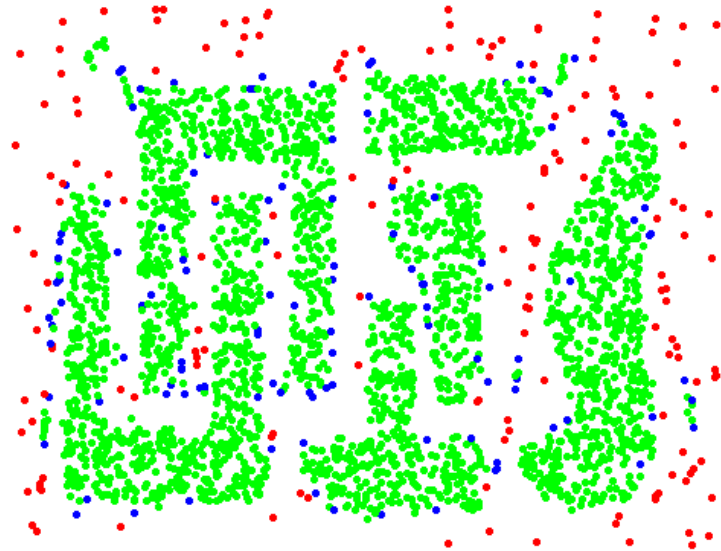
Another Version of DBSCAN (for simplicity, not efficiency)

- ▶ Label all points as core, border or noise points
- ▶ Eliminate noise points
- ▶ Put an edge between all core points that are within *Eps* of each other
- ▶ Make each group of connected core points into a separate cluster
- ▶ Assign each border point to one of the clusters of its associated core points.

DBSCAN: Core, Border and Noise Points



Original Points

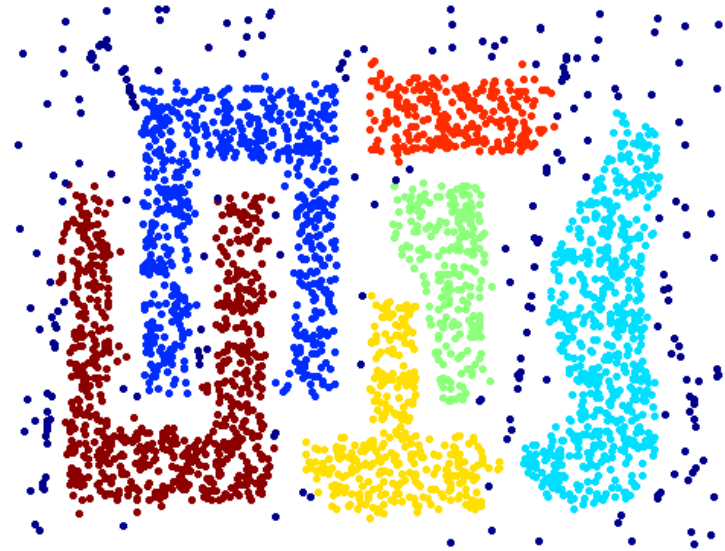


Point types: **core**,
border and **noise**

When DBSCAN Works Well



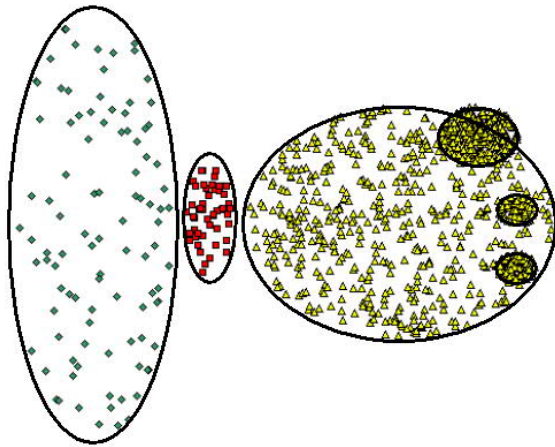
Original Points



Clusters

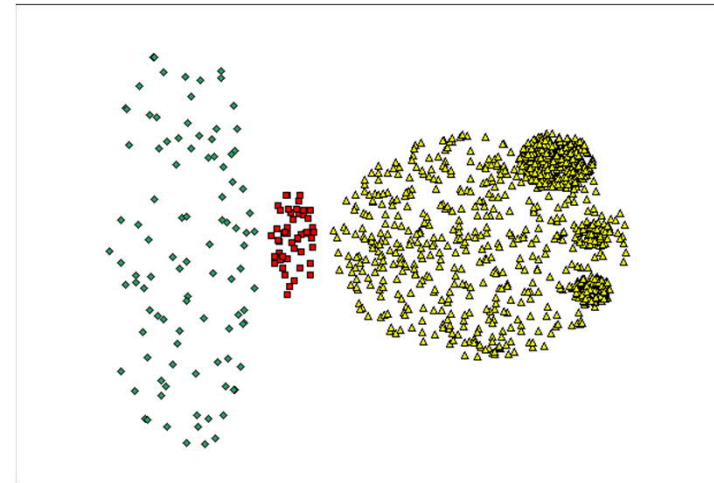
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN Does NOT Work Well

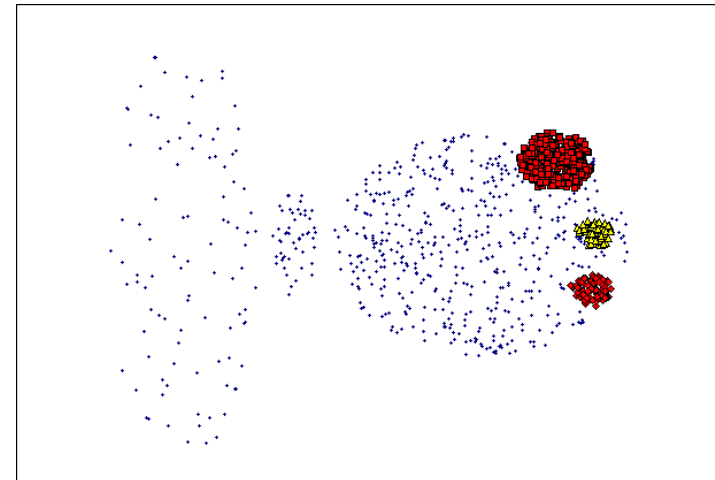


Original Points

- Varying densities



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

DBSCAN: Sensitive to Parameters

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

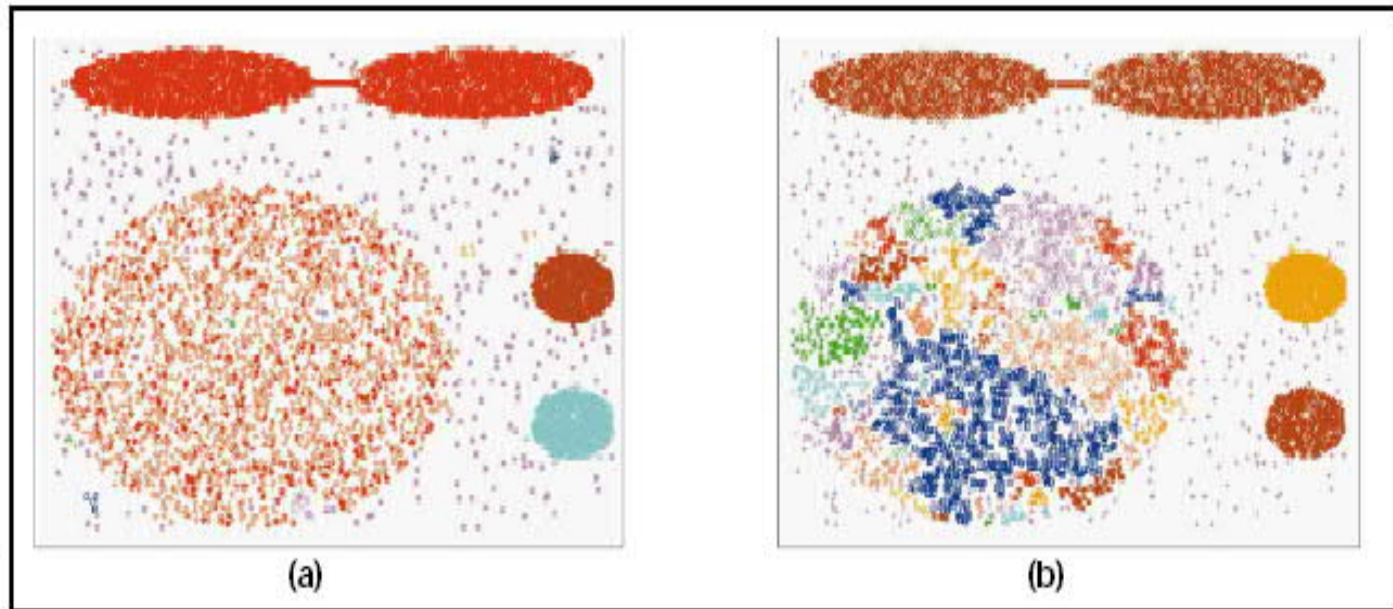
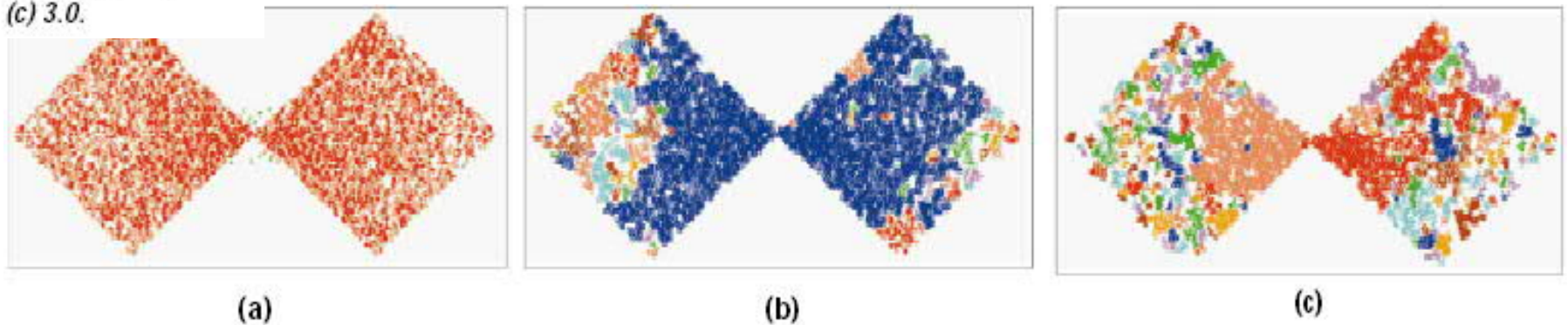


Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.



Comparing DBSCAN and K-means

- ▶ Cluster shapes

- ▶ K-means: spheres
- ▶ DBSCAN: arbitrary shape



- ▶ Cluster sizes

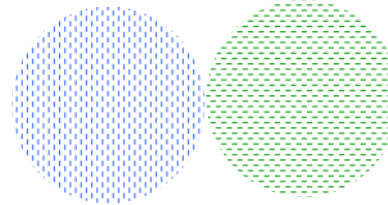
- ▶ K-means may have a problem when clusters are of different sizes and are close to each other.
- ▶ DBSCAN can handle clusters of different sizes

- ▶ Noise and outliers

- ▶ K-means is sensitive to noise or outliers
- ▶ DBSCAN is not strongly affected by noise or outliers

Comparing DBSCAN and K-means (Cont'd)

- ▶ Not well-separated clusters

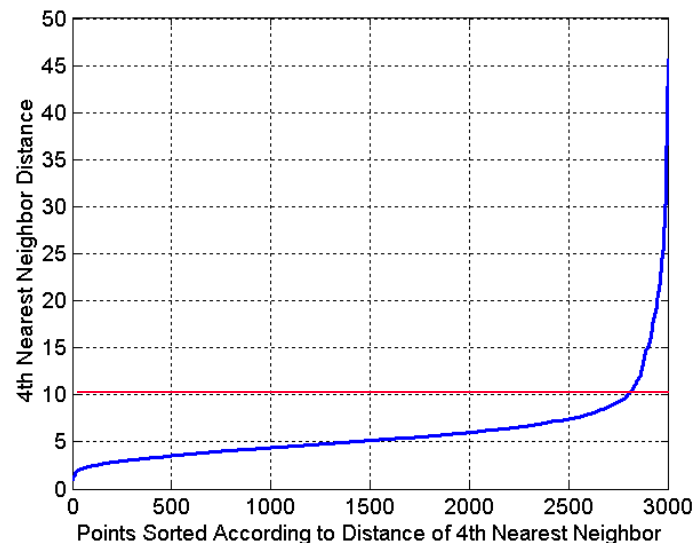


- ▶ K-means can find clusters that are not well separated, even if they overlap
 - ▶ DBSCAN merges clusters that overlap
- ▶ Stability
 - ▶ K-means' result depends on the random initialization of centroids
 - ▶ DBSCAN produces the same set of clusters from one run to another (except that the membership of some border points depends on the order of the points being processed)
- ▶ Number of clusters
 - ▶ For k-means, the number of clusters needs to be specified as a parameter
 - ▶ DBSCAN automatically determines the number of clusters. However, it has two other parameters: *Eps* and *MinPts*

DBSCAN: Determining Eps Given MinPts

- ▶ Let $\text{MinPts} = k$
- ▶ Idea is that for points in a cluster, their k^{th} nearest neighbors are at roughly the same distance
- ▶ Noise points have the k^{th} nearest neighbor at farther distance
- ▶ So, plot sorted distance of every point to its k^{th} nearest neighbor (e.g., $k=4$)

Thus, $\text{Eps}=10$ for $\text{MinPts}=4$



Comparing to Hierarchical Clustering

► Key differences

Feature	DBSCAN	Hierarchical Method
Cluster Shape	Arbitrary (work well for non-convex shapes)	Depends on linkage
Noise Handling	Explicitly identifies noise	No inherent noise handling
Scalability	$O(n \log n)$ with spatial indexing	$O(n^3)$ or $O(n^2 \log n)$
Parameters	Eps, MinPts	Number of clusters or distance threshold
Hierarchy	No hierarchy	Provides a dendrogram
Number of Clusters	Automatic	Specified (if not using distance threshold)

Pros and Cons of DBSCAN

▶ **Pros:**

- ▶ Handles noise and outliers well
- ▶ Discovers clusters of arbitrary shapes
- ▶ No need to predefine the number of clusters

▶ **Cons:**

- ▶ Struggles with varying densities
- ▶ Sensitive to Eps and MinPts

Outline

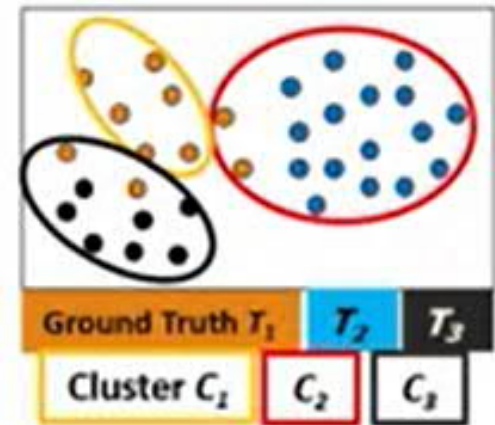
- ▶ What is Clustering?
- ▶ Types of Data to Cluster
- ▶ Distance/ Similarity Measures
- ▶ Clustering Methods
 - ▶ K-means and K-means++
 - ▶ K-medoids
 - ▶ Hierarchical clustering method
 - ▶ DBSCAN
- ▶ Cluster Validity Measures

Measures of Clustering Quality

- ▶ Two types of numerical measures:
 - ▶ **External Index:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - ▶ The true class labels are *external* to the data set
 - ▶ E.g., Rand statistic, Jaccard, entropy
 - ▶ **Internal Index:** Used to measure the goodness of a clustering structure *without* external information.
 - ▶ E.g., Sum of Squared Error (SSE), Davies-Bouldin (DB) index
- ▶ These are also referred to as **criteria**
 - ▶ However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.

External Indexes

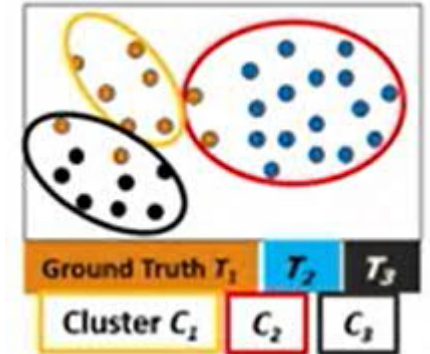
- ▶ Ground truth is given
 - ▶ *True partitions/classes of the data points are known.*
 - ▶ Clustering result is compared with the true partitions



True Positives, True Negatives, False Positives and False Negatives

▶ True Positives

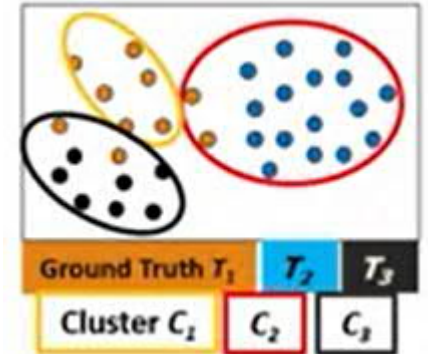
- ▶ x_i and x_j are a *true positive pair* if they belong to the same partition and are also in the same cluster
- ▶ TP is the number of true positive pairs



▶ False Positives

- ▶ x_i and x_j are a *false positive pair* if they do not belong to the same partition, but are clustered into the same cluster
- ▶ FP is the number of false positive pairs

True Positives, True Negatives, False Positives and False Negatives



▶ True Negative

- ▶ x_i and x_j are a *true negative pair* if they do not belong to the same partition and are also not in the same cluster
- ▶ TN is the number of true negative pairs

▶ False Negative

- ▶ x_i and x_j are a *false negative pair* if they belong to the same partition, but are clustered into different clusters
- ▶ FN is the number of false negative pairs

Rand Statistic/Index

- ▶ Rand Statistic/Index (also called HA Index)
 - ▶ Measures the fraction of true positives and true negatives over all pairs

$$Rand = \frac{TP + TN}{TP + FP + TN + FN}$$

- ▶ Measures the fraction of point pairs where both clustering result and the ground truth agree.
 - ▶ A perfect clustering has a value of 1 for the statistic.

Jaccard Coefficient

- ▶ Measures the fraction of true positive point pairs, but after ignoring the true negatives:

$$Jaccard = \frac{TP}{TP + FP + FN}$$

- ▶ For a perfect clustering, the coefficient is 1, that is, there are no false positives nor false negatives

Internal Indexes

- ▶ Based on data only, without consulting *external* true partitions
- ▶ Normally consider intra-cluster similarity or **compactness** contrasted with the notations of inter-cluster **separation**
- ▶ Typically propose a trade-off to maximizing these two competing measures.

SSE: an Internal Index

- ▶ Sum of Squared Error (SSE):

$$SSE = \sum_{i=1}^k \sum_{x \in C_i} \text{dist}(x, v_i)^2$$

v_i is the center of the cluster C_i

k is the number of clusters

- ▶ Considers only the compactness (i.e., the intra-cluster distances) of the clusters

DB: an Internal Index

- ▶ Davies-Bouldin index (DB) :

$$DB = \frac{1}{k} \sum_i \max_{j=1..k, j \neq i} (d_{ij}) \quad \text{where} \quad d_{ij} = \frac{\sigma_i + \sigma_j}{d(v_i, v_j)}$$

- ▶ k is the number of clusters,
- ▶ σ_i is the average distance between cluster points and the center in the i th cluster
- ▶ $d(v_i, v_j)$ is the distance between the i th and j th cluster centers.
- ▶ d_{ij} decreases when the clusters are more compact and when the distance between them is larger.
- ▶ For each cluster C_i , the $\max(d_{ij})$ identifies its “worst” neighbour. The DB index is the average of such value for all clusters.
- ▶ The DB index varies on the interval $[0, \infty)$ and is *small* when the clusters are *compact and well separated*.

Final Comment on Cluster Validity

“The validation of clustering structures is the most difficult and frustrating part of cluster analysis.

Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage.”

Algorithms for Clustering Data, Jain and Dubes

Summary

- ▶ What is Clustering?
- ▶ Types of Data to Cluster
- ▶ Distance/ Similarity Measures
- ▶ Clustering Methods
 - ▶ K-means and K-means++
 - ▶ K-medoids
 - ▶ Hierarchical clustering method
 - ▶ DBSCAN
- ▶ Cluster Validity Measures